SWARMOPS

AUTONOMOUS AI ENGINEERING TEAM

One objective. Six agents. One engineered solution.

Open the live application

Watch the product walkthrough

GitHub repository: [ADD GITHUB REPOSITORY URL]

Project specification • Prepared 8 November 2026

1. Executive summary

SwarmOps is a multi-agent engineering workspace that turns a natural-language software objective into a coordinated workflow across research, architecture, development, testing, security review, and delivery. A central orchestrator coordinates specialist roles and consolidates their work into a traceable engineering handoff.

2. Problem statement

Software delivery spans several disciplines and tools. Research, architecture, implementation, quality assurance, security, and release documentation are often disconnected. A single AI response can be plausible without providing a transparent account of specialist contributions, dependencies, validation evidence, or unresolved risks.

3. Product solution

SwarmOps presents engineering work as a coordinated team process. The user submits an objective, the orchestrator creates tasks, agents contribute role-specific outputs, review findings can be routed back for revision, and the Delivery Agent assembles the available outputs.

4. Six-agent operating model

Agent	Primary responsibility	Expected outputs
Research	Requirements, constraints, technology investigation	Research brief, assumptions, options, sources
Architect	System structure, interfaces, data flows	Architecture, API/data design, implementation plan
Developer	Implementation of approved design	Source files, project structure, README
Tester	Validation planning and available checks	Test plan, results, defects, regression status
Security	Security review of design and code	Findings, severity, remediation guidance
Delivery	Consolidation and release handoff	Final report, deployment guide, release checklist

5. End-to-end workflow

• User submits objective and optional technology preferences.

• Orchestrator decomposes the objective into tasks and dependencies.

• Research findings inform the architecture.

• Developer implements the approved design.

• Tester and Security review available implementation artifacts.

• Findings are routed back for remediation where supported.

• Delivery consolidates artifacts, evidence, and known limitations.

6. Application experience

Command Center

The published landing page identifies the product as “Autonomous AI Engineering Team” and uses the statement “One objective. Six agents. One engineered solution.” It provides an objective input, project name, optional technology preference, Start Swarm (Live AI), Run Demo, and example objectives.

Execution modes

Demo Mode: a scripted scenario for repeatable presentations. Simulated events must be labelled as simulated. **Live AI Mode:** the application page states that it runs six real AI agents on the submitted objective and may take a few minutes.

Execution boundary

The app explicitly states that code is reviewed but not executed in either mode. Therefore, generated code must not be described as runtime-tested or deployment-ready solely because an agent reviewed it.

7. Example use case: EV Charging Management API

Objective: build a secure API for an electric vehicle charging platform managing users, charging stations, charging sessions, energy consumption, and reporting. Research defines requirements and candidate technologies; Architecture defines services and data contracts; Development produces implementation artifacts; Testing and Security review available outputs; Delivery prepares the engineering handoff. This is an illustrative scenario, not a claim about the outcome of a particular run.

8. Orchestration and collaboration requirements

• Task ownership, dependencies, status, timestamps, and structured outputs.

• Agent-to-agent messages with sender, recipient, context, and artifact references.

• Bounded retries, timeouts, cancellation, and error handling.

• Shared project context with traceable source artifacts.

• Human approval before deployment, destructive operations, or external changes.

• Clear distinction between generated, reviewed, executed, and verified work.

9. Suggested reference architecture

Web client: objective entry, command center, agent views, timeline, artifact viewer. **Orchestrator:** task decomposition, dependency management, scheduling, retries, cancellation. **Agent workers:** role-specific prompts, tools, structured outputs, bounded permissions. **Persistence:** projects, runs, tasks, messages, events, artifacts, test results, findings. **Model gateway:** provider abstraction and server-side secrets. **Optional sandbox:** isolated builds and tests with resource and network restrictions. **Artifact service:** reports, versioning, storage, downloadable bundles.

10. Data entities

Suggested entities: Project, Objective, ExecutionRun, Agent, AgentTask, AgentMessage, ExecutionEvent, Artifact, TestResult, SecurityFinding, and Approval. Every record should be scoped to its project and execution run, with authorization enforced server-side.

11. Security, reliability, and trust

• Keep model-provider keys on the server; never ship secrets to browser code.

• Apply project-level access control and audit important actions.

• Do not execute generated code on the main application server or in the browser.

• Use an isolated execution environment if runtime validation is added.

• Attach evidence and severity to security findings.

• Label simulated results and distinguish automated checks from manual review.

• Require human approval for high-impact external actions.

12. Demo walkthrough

• Open the live application: https://agents-swarm-ops.lovable.app/

• Select Run Demo for a scripted walkthrough, or Start Swarm (Live AI) for a real agent run.

• Use the EV Charging API example objective.

• Show role specialization, coordination, review feedback, and delivery.

• Explain that the published app currently describes code as reviewed but not executed.

• Play the video walkthrough: https://youtu.be/iNc_CAn4aFs

13. Scope and verification notes

This specification is grounded in the published application landing page and the links supplied by the project owner. The app page was readable, but an actual browser screenshot could not be captured in this environment. The PowerPoint therefore uses clearly labelled illustrative UI mockups rather than misrepresenting them as live screenshots. Internal backend implementation details should be confirmed against the source repository.

14. Project links

Application: https://agents-swarm-ops.lovable.app/
Demo: https://youtu.be/iNc_CAn4aFs
GitHub: [ADD GITHUB REPOSITORY URL] (replace with actual repository URL)